

FINAL PROJECT ASSIGNMENT

HTML, CSS & JavaScript

Project: Student Task & Productivity Dashboard

Objective: Build a functional, responsive productivity dashboard that allows users to create, manage, search, filter and track tasks using HTML, CSS and vanilla JavaScript.

1. Project Overview

Students will build a real-world task management application. The project should demonstrate their ability to combine HTML structure, CSS styling and JavaScript logic to create an interactive web application.

2. Required Features

Feature	Requirement
Dashboard	Display total tasks, completed tasks and pending tasks. The figures must update automatically.
Add Task	Provide a form for task title, description, due date and priority. Validate the form and add the task dynamically.
Task Management	Users must be able to mark tasks as completed, edit tasks and delete tasks.
Search	Provide a search field that dynamically filters tasks based on the task title or relevant text.
Filter	Provide filters for All, Pending, Completed and High Priority tasks.
Local Storage	Tasks must remain available after refreshing or closing and reopening the browser by using localStorage.
Productivity Statistics	Display completed, pending and completion percentage. The percentage must be calculated dynamically.
Dark/Light Mode	Allow users to switch between light and dark themes and remember their selected theme.

3. Example Task Card

Each task should present useful information such as the title, description, due date and priority, along with controls for Complete, Edit and Delete.

4. JavaScript Concepts Expected

- Variables and data types
- Functions
- Arrays and objects
- DOM manipulation
- Event handling
- Conditional statements
- Array methods such as map(), filter(), find() and reduce()
- Form validation

- localStorage
- Date handling

5. Extra Challenge Features

- Sort tasks by due date, priority, newest or oldest.
- Detect overdue tasks automatically.
- Display a confirmation before deleting a task.
- Add smooth UI animations and transitions.
- Make the application fully responsive on mobile, tablet and desktop.

6. Project Rules

- Use HTML, CSS and vanilla JavaScript.
- Google Fonts, Font Awesome or Lucide icons may be used.
- Do not use React, Bootstrap, Tailwind CSS, jQuery or other JavaScript frameworks.
- Do not use ready-made dashboard templates.
- The project must be built from scratch.
- Students must be able to explain their HTML, CSS and JavaScript during presentation.

7. Marking Scheme — 100 Marks

Assessment Area	Marks
HTML structure & semantics	10
CSS/UI design	15
Responsive design	10
Add task functionality	10
Edit/Delete functionality	10
Complete/Pending functionality	10
Search & filtering	10
LocalStorage	10
Statistics/calculations	5
Dark/Light mode	5
Code quality	5
TOTAL	100

Bonus: +10 marks

Award bonus marks for advanced features such as sorting, overdue detection, animations and excellent UI/UX.

8. Project Presentation

Students should demonstrate the completed application and explain how the JavaScript logic works. The instructor may ask questions about how tasks are stored, filtered, updated, deleted and persisted.

- Where are the tasks stored?
- Why did you use filter() here?
- What happens when the page is refreshed?
- How is the task status changed?
- How is the completion percentage calculated?
- What happens if the user submits an empty form?

9. Expected Outcome

By the end of the project, students should have a complete, responsive and functional productivity dashboard that demonstrates practical understanding of HTML, CSS and JavaScript.